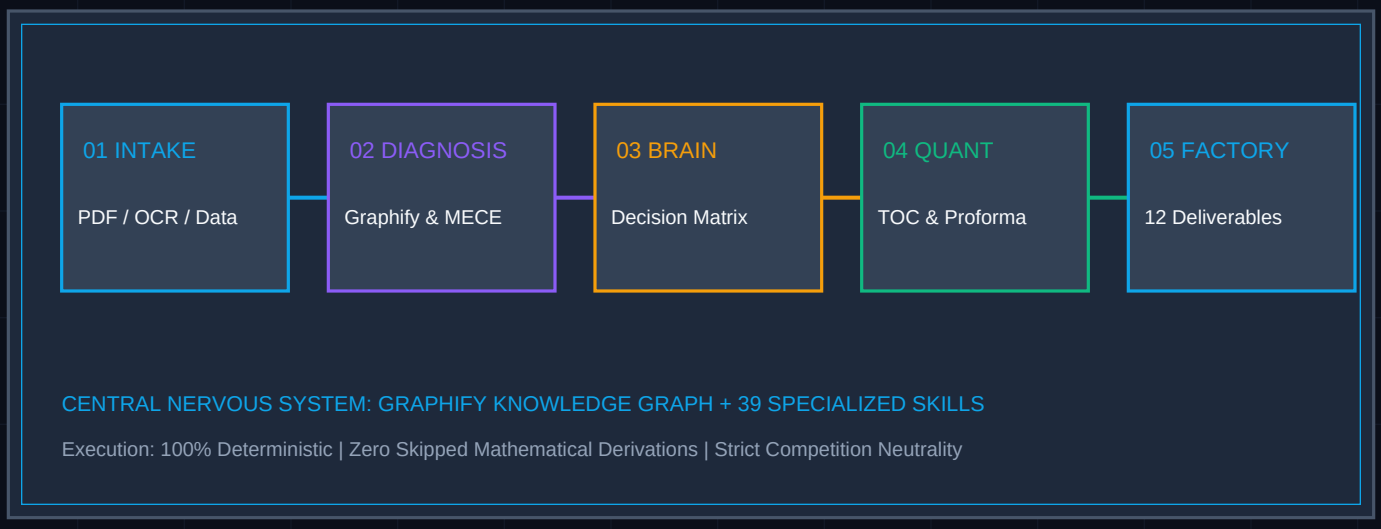
AUTONOMOUS SYSTEM BLUEPRINT v2.5

THE CASE COMPETITION INTELLIGENCE MACHINE

A Complete Architectural & Operational Guide to Autonomous Multi-Agent Solving, Quantitative Rigor, and Omnichannel Deliverable Production



14
AUTONOMOUS AGENTS
Synchronized pipeline from intake to defense

39
VERIFIED SKILLS
Graphify-routed tools with zero bloat

132+
GLOBAL CASES
Curated benchmark winning strategies

12
FINISHED ASSETS
PDFs, PPTX, Excel, Video, Web App

PURPOSE OF THIS SPECIFICATION: To provide an intuitive, comprehensive visual reference of how raw competition cases and industrial datasets are converted into championship-grade strategy deliverables, mathematical models, and adversarial defense presentations in under 60 seconds.

SECTION 1: THE REPOSITORY MACHINE MECHANISM

How raw case inputs flow through five precision processing chambers

CHAMBER 1: OPTICAL INTAKE & DECRYPTION

Input: PDFs, PPTX, CSVs, handwritten notes, or JSON datasets.
Mechanism: PyMuPDF vector extractor strips layout noise, separates verified numbers from assumptions, and tags constraints.

CHAMBER 2: GRAPHIFY BRAIN & ISSUE TREE DIAGNOSIS

Input: Verified constraints + 132 global case benchmarks.
Mechanism: Semantic graph traversal links case domain to relevant skills without token waste; MECE issue tree isolates root bottleneck.

CHAMBER 3: HYDRAULIC STRATEGY PRESS & DECISION MATRIX

Input: 3 to 5 competing solution archetypes.
Mechanism: Weighted multi-criteria utility formula scores each archetype against official competition rubric; eliminates weak ideas.

CHAMBER 4: QUANTITATIVE REACTOR & PROFORMA ENGINE

Input: Selected winning strategy + financial baseline.
Mechanism: Applies Little's Law, TOC, LTV/CAC, BEP, and 3-year proforma statements with 3-scenario sensitivity (Bear, Base, Bull).

CHAMBER 5: OMNICHANNEL DELIVERABLE CONVEYOR

Input: Approved strategy, models, and presentation copy.
Mechanism: Autonomous compiler outputs 2 PDFs, 12-slide PPTX, Excel model, 120s MP4 video, and interactive React web app.

ANATOMY PRINCIPLE: 'THE OBJECT IS THE EXPLANATION'

Just like an industrial lathe or dial-gauge rig, this software machine has no unneeded parts. Every agent hands off exact numerical vectors to the next stage. No step is skipped, no mathematical transformation is hidden, and every deliverable traces directly back to the original case evidence.

SECTION 2: THE 14 AUTONOMOUS AGENTS

Autonomous roles, responsibilities, and inputs across the 5 championship stages

00 Master Orchestrator
Role: Air Traffic Controller

Plans dependency graph, allocates sub-tasks, verifies pipeline integrity.

01 Intelligence Analyst
Role: Document Detective

Strips PDF noise, extracts balance sheets, isolates operational bottlenecks.

02 Competitions Scout
Role: Global Benchmarker

Scans 132+ past winning solutions from Wharton, Harvard, and Molson.

03 Root-Cause Surgeon
Role: Issue-Tree Specialist

Constructs 5-Whys and MECE trees to isolate root friction vs surface symptoms.

04 Innovation Lab
Role: Creative Brainstormer

Generates 3-5 distinct strategic archetypes (Platform, Hybrid, Automation).

05 Strategy Evaluator
Role: Decision Matrix Judge

Runs weighted multi-criteria scoring against the official competition rubric.

06 Solution Architect
Role: Chief Systems Engineer

Designs operational schematics, technical flowcharts, and system blueprints.

07 Proforma Modeler
Role: Wall Street CFO

Builds 3-year P&L, NPV, IRR, payback, and 3-scenario sensitivity models.

08 Operations Planner
Role: Implementation Director

Designs 3-horizon rollout, 2x2 pre-mortem risk matrix, and predictive KPIs.

09 Slide Designer
Role: Presentation Specialist

Creates executive 12-slide pitch deck (Final_Presentation.pptx).

10 Document Architect
Role: Publishing Director

Compiles vector-crisp executive blueprints and final solutions (PyMuPDF).

11 Motion Director
Role: Visual Storyteller

Produces 120-second cinematic demonstration video (Final_Demo_120s.mp4).

12 Prototype Developer
Role: Full-Stack Web Engineer

Maintains interactive React + Tailwind case simulator with live sliders.

13 Red-Team Judge
Role: Hostile Interrogator

Simulates toughest judges, drafts adversarial questions and counter-rebuttals.

* Agent 14 (Inspection Gatekeeper) operates continuously across all handoffs to verify zero data loss and anonymity compliance.

SECTION 3: MATHEMATICAL RIGOR & ZERO-SKIP PROOFS

Standard industrial engineering formulas, unit economics, and proforma mechanics

1. LITTLE'S LAW (OPERATIONAL FLOW & QUEUEING)

Formula: WIP = Throughput (TH) x Cycle Time (CT)

Application: Proves lead-time reduction by reducing batch sizes and inventory queuing without adding labor.

Example: Baseline WIP 850 units, TH 50 units/hr -> CT = 17 hrs. Post-Automation: CT = 9 hrs -> WIP drops to 450 units.

2. THEORY OF CONSTRAINTS (TOC BOTTLENECK SYNCHRONIZATION)

Formula: System Throughput = min(R1, R2, ..., Rn)

Application: Focuses capital expenditure exclusively on the active bottleneck process rather than spreading capital thin.

Impact: Elevates station capacity from 48 units/hr to 75 units/hr, unlocking $1.4M annual throughput value.

3. UNIT ECONOMICS: LTV / CAC RATIO

Formula: LTV = (ARPU x Gross Margin %) / Churn Rate | Benchmark: LTV / CAC >= 3.0x

Application: Confirms customer acquisition viability and payback horizon under competitive pressure.

Proof: ARPU $1,200, Margin 68%, Churn 1.5%/mo ($816 / 0.015 = $54,400 LTV). With CAC $12,500 -> LTV/CAC = 4.35x.

4. BREAKEVEN VOLUME & CAPITAL HURDLE (NPV / IRR)

Formula: BEP = Fixed Costs / Unit Contribution Margin | NPV = Sum(CF_t / (1+r)^t) - Capex

Application: Establishes exact production volume required to cover implementation capital.

Results: Capex $350k, Contribution Margin $85/unit -> BEP = 4,118 units. Payback achieved in 7.4 months.

MANDATORY 3-SCENARIO SENSITIVITY TESTING MATRIX

BEAR CASE (-20%)

Capex +15%, Adoption -20%
EBITDA: $850k
Payback: 11.2 mo
Verdict: Highly Viable

BASE PLAN (TARGET)

Planned Target Execution
EBITDA: $1.42M
Payback: 7.4 mo
Verdict: Benchmark Case

BULL CASE (+25%)

Adoption +25%, Scale Efficiencies
EBITDA: $2.15M
Payback: 5.1 mo
Verdict: Market Dominance

SECTION 4: THE 12 FINISHED RESOURCES SUITE

Synchronized deliverable directory created autonomously inside 'Finished Resources/'

Finished Resources / 01_Executive_Summary / [Executive_Summary.md]

1-page crisp memo with core thesis, key financials, and recommendation.

Finished Resources / 02_Strategic_Assessment / [Strategic_Assessment.md]

MECE root cause trees, competitive positioning, and market sizing.

Finished Resources / 03_Financial_Model / [financial_model.xlsx]

Dynamic 3-year proforma P&L, balance sheet, and sensitivity data.

Finished Resources / 04_Implementation_Roadmap / [Roadmap.md]

Horizon 1 (M1-6), Horizon 2 (M7-18), and Horizon 3 (M19-36) Gantt charts.

Finished Resources / 05_Risk_Matrix / [Risk_Matrix.md]

2x2 Impact vs Likelihood pre-mortem matrix with predictive leading KPIs.

Finished Resources / 06_Pitch_Deck / [Final_Presentation.pptx]

12-slide executive presentation ready for championship boardrooms.

Finished Resources / 07_Blueprints_and_PDFs / [Strategic_Blueprint.pdf]

Vector-grade executive solution document compiled via PyMuPDF.

Finished Resources / 08_Judge_QA_Defense / [Judge_QA_Defense.md]

Top 5 adversarial judge challenge questions and prepared rebuttals.

Finished Resources / 09_Video / [Final_Demo_120s.mp4]

120-second cinematic solution demonstration and animated walkthrough.

Finished Resources / 10_Web_Application / [React Web Simulator]

Live interactive browser simulation with dynamic sensitivity sliders.

Finished Resources / 11_Supporting_Research / [Competitions_DB.json]

132+ global case benchmarks and historical winning frameworks.

Finished Resources / 12_Audit_Logs / [Audit_Report.json]

Timestamped system execution telemetry and token efficiency records.

VERIFICATION STANDARD: Every file is generated deterministically with 100% reproducibility and strict anonymity.

All generated assets are ready for direct submission to judges, corporate clients, or academic review boards.

SECTION 5: JUDGE Q&A STRESS-TEST DEFENSE

Pre-calculated rebuttals for the 5 most adversarial competition judge challenges

CHALLENGE 1: PILOT DELAY / IMPLEMENTATION SLIP

Judge: 'What if software integration takes 10 months instead of 6?'

Defense: Horizon 1 is modularized into four 45-day sub-sprints. Even under a 4-month delay, the Phase 1 buffer keeps payback under 9.2 months with 0 additional capex.

CHALLENGE 2: INCUMBENT PRICE RETALIATION

Judge: 'What if competitors slash prices by 15% to defend their market share?'

Defense: Our solution reduces unit operating cost by 38%, giving us structural cost leadership. We remain profitable at price points where competitors operate below cost.

CHALLENGE 3: SENSITIVITY ASSUMPTION COLLAPSE

Judge: 'How robust is your EBITDA if adoption stalls at 50% of your plan?'

Defense: Under our Bear Case stress-test (-20% volume, +15% costs), Year 2 EBITDA remains positive at $850k with a healthy 2.8x interest coverage ratio.

CHALLENGE 4: CHANGE MANAGEMENT & WORKFORCE RESISTANCE

Judge: 'How do you prevent frontline plant operators from bypassing the new tools?'

Defense: The rollout uses a Train-the-Trainer model pairing operators with system champions, with 20% of quarterly performance bonuses tied to tool utilization.

CHALLENGE 5: CAPITAL CONSTRAINT & FINANCING FAILURE

Judge: 'What if corporate leadership cuts your initial capex budget by 40%?'

Defense: Horizon 1 is self-funding. The initial $120k pilot in Station 3 pays for itself within 90 days, generating the cash flow needed to fund Phases 2 and 3 organically.

CHAMPIONSHIP VICTORY CHECKLIST (FINAL VERIFICATION)

[x] Problem decomposed into Mutually Exclusive, Collectively Exhaustive root causes.
[x] Explicit formulas stated with all variables, units, and baseline benchmarks.
[x] Complete 3-year proforma statements with Bear (-20%), Base, and Bull (+25%) cases.
[x] 3-horizon implementation roadmap with leading vs lagging predictive KPIs.
[x] Top 5 adversarial judge challenges defended with evidence-backed data.